THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 06

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** time taken to puff up	A number in seconds
2	**(c)** a place for scientific observation	Science is everywhere
3	**(d)** almost everywhere, even inside other organisms	Found in all environments
4	**(d)** non-communicable diseases	Lifestyle changes
5	**(a)** sign	Observed or measured
6	**(d)** copper wire and iron nail	Two different metals
7	**(d)** navigate	They sense the field
8	**(d)** newtons	A force
9	**(b)** pressure inside our body balances it	Internal pressure equal to outside
10	**(a)** move towards each other	Low pressure between
11	**(d)** evaporation	A surface process
12	**(b)** cannot be broken down into simpler substances	Building blocks of matter
13	**(a)** natural solid substances with fixed chemical composition	Found in Earth
14	**(d)** warm water	Oxygen solubility decreases
15	**(a)** converge	Converging mirror
16	**(a)** reflects sunlight	Non-luminous
17	**(d)** 1987	As per the chapter
18	**(a)** both organisms benefit	e.g. flower and bee
19	**(d)** two parents and fusion of gametes	Variation arises
20	**(d)** greenhouse gases like carbon dioxide	Raises Earth's temperature

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Microorganisms that break down dead plants and animals (1). Bacteria or fungi (1).	1+1
22	A scientist who studied bacteria (1). He discovered penicillin from a mould in 1928 (1).	1+1
23	Zinc container (negative terminal) (1). Carbon rod with metal cap (positive) in paste electrolyte (1).	1+1
24	Lightning heats air rapidly (1). Air expands suddenly and makes a loud sound (1).	1+1
25	Minerals that are pure elements (1). Example: gold (1).	1+1
26	Balance measures mass (1). Cylinder measures volume of liquids (1).	1+1
27	Growing the same crop repeatedly (1). Reduces biodiversity and affects pollinators (1).	1+1
28	To avoid competition for resources (1). To grow in new places (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Use identical dough circles (1). Drop one vertically and one at an angle in the same oil (1). Compare puffing and time (1).	1+1+1
30	Ship displaces much water (1). Upthrust equals its weight (1). Nail displaces little water so sinks (1).	1+1+1
31	Particles vibrate more (1½). They move slightly apart, increasing size (1½).	1½+1½
32	Concave lens (1½). It diverges rays so image falls properly on the retina (1½).	1½+1½
33	Researcher in space science and nuclear physics (1). Father of Indian Space programme (1). VSSC is named after him (1).	1+1+1

34	Many species perform different roles (1½). If one is lost, others keep balance (1½).	1½+1½
35	It becomes very hot (1½). It can cause burns (1½).	1½+1½
	OR: Use two different metal electrodes in a lemon (1). The juice works as electrolyte (1). A chemical reaction produces small current (1).	1+1+1
36	Natural: forest, pond (1½). Human-made: farm, fish pond (1½).	1½+1½
	OR: Tigers eat deer and wild boar (1). If prey declines, tigers starve (1). Needs healthy forests (1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	Diet (1). Activity (1). Sleep (1). Environment (1). Mental wellbeing (1).	1+1+1+1+1
	OR: Definition (1). Physical example (1). Mental example (1). Social example (1). Two habits such as balanced diet and exercise (1).	1+1+1+1+1
38	High to low pressure principle (1). Day: sea breeze (1½). Night: land breeze (1½). Diagram labelled (1).	1+1½+1½+1
	OR: Pressure = F/A (1). More force gives more pressure (1). Less area gives more pressure e.g. knife (1½). More area less pressure e.g. broad strap (1½).	1+1+1½+1½
39	Temperature rise lowers density (1). Example – hot air rises (1). Pressure raises gas density (1). Liquids and solids change little (1). Example – compressed gas cylinder (1).	1+1+1+1+1
	OR: Ice particles are farther apart (1). Ice has lower density (1). Floats and forms top layer (1). Insulates water beneath (1). Aquatic life survives (1).	1+1+1+1+1
40	Farms, fish ponds (1). Pollination (1). Pest control (1). Soil nutrients (1). Water retention (1).	1+1+1+1+1
	OR: Decomposers feed on dead matter (1). Break complex matter into simpler (1). Compost forms (1). Nutrients return to soil (1). Cycle continues (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER KEY